

Modular Operator Discovery TSP Report

Overview

- Condition count: 7.
- Best final transfer gap: phase7_full_solver_evolution.
- Best held-out TSPLIB gap: phase7_full_solver_evolution.
- Surviving modular candidates: none.

Run Metadata

- run_name: run_20260520_045743_m
- started_at_local: 2026-05-20 04:57:43
- finished_at_local: 2026-05-20 05:32:04
- duration_hhmm: 00:34
- duration_seconds: 2060.508
- seed_offset: 12000
- replicate_label: m
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_baseline_heuristic_only	baseline_only	none	score_only	0.235058	0.054968	0.121317	0.0	0.0	False	0.0	0.0
phase7_full_solver_evolution	full_solver	none	score_only	0.180847	0.0	0.115084	0.78157	0.76	False	0.0	0.0
phase7_modular_operator_evolution	modular_operator	none	score_only	0.235058	0.054968	0.121317	0.865478	0.445	False	0.003002	-0.000838
phase7_modular_operator_random_replay	modular_operator	random	score_only	0.235667	0.054968	0.121542	0.875576	0.4425	False	0.010654	0.610699
phase7_modular_operator_diversity_residual_replay	modular_operator	diversity_residual	score_only	0.235058	0.049737	0.118013	0.874239	0.4425	False	0.007732	0.623816

Condition	Mode	Replay	Selection	Final TSPLIB Gap	Final Family Gap	Final Transfer Gap	Mean Novelty	Mean Complexity	Surviving Candidate	Transplant Delta	Pareto Runtime Inflation
phase7_modular_operator_compression_pressure	modular_operator	none	novelty_gate	0.235058	0.050827	0.118702	0.0	0.4425	False	0.00714	0.616565
phase7_modular_operator_pareto_selection	modular_operator	none	pareto	0.235667	0.054968	0.121542	0.795501	0.44	False	0.003293	0.197399

Condition Notes

phase7_baseline_heuristic_only

- Execution mode baseline_only on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.0, complexity 0.0, and adaptation efficiency 0.0.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_full_solver_evolution

- Execution mode full_solver on host scaffold whole_solver.
- Final gaps: TSPLIB 0.180847, family holdout 0.0, combined 0.115084.
- Accepted novelty 0.78157, complexity 0.76, and adaptation efficiency 0.014874.
- Last-epoch runtime 0.0 ms and distance evaluations 0.0.
- Validation: surviving False, transplant delta 0.0, positive scaffolds 0, Pareto runtime inflation 0.0.

phase7_modular_operator_evolution

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.054968, combined 0.121317.
- Accepted novelty 0.865478, complexity 0.445, and adaptation efficiency 0.004584.
- Last-epoch runtime 2071.520475 ms and distance evaluations 2755497.0.
- Validation: surviving False, transplant delta 0.003002, positive scaffolds 0, Pareto runtime inflation -0.000838.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.0, "gap_delta": 0.0, "runtime_inflation": -0.000838, "same_gap_faster": true}

phase7_modular_operator_random_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.875576, complexity 0.4425, and adaptation efficiency 0.007846.
- Last-epoch runtime 257.032825 ms and distance evaluations 10482.75.

- Validation: surviving False, transplant delta 0.010654, positive scaffolds 1, Pareto runtime inflation 0.610699.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.433839, "gap_delta": 0.000224, "runtime_inflation": 0.610699, "same_gap_faster": false}

phase7_modular_operator_diversity_residual_replay

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.049737, combined 0.118013.
- Accepted novelty 0.874239, complexity 0.4425, and adaptation efficiency 0.017392.
- Last-epoch runtime 59.9857 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.007732, positive scaffolds 1, Pareto runtime inflation 0.623816.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.763082, "gap_delta": -0.005462, "runtime_inflation": 0.623816, "same_gap_faster": false}

phase7_modular_operator_compression_pressure

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235058, family holdout 0.050827, combined 0.118702.
- Accepted novelty 0.0, complexity 0.4425, and adaptation efficiency 0.0.
- Last-epoch runtime 56.66135 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.00714, positive scaffolds 1, Pareto runtime inflation 0.616565.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.763692, "gap_delta": -0.003304, "runtime_inflation": 0.616565, "same_gap_faster": false}

phase7_modular_operator_pareto_selection

- Execution mode modular_operator on host scaffold nearest_neighbor_2opt.
- Final gaps: TSPLIB 0.235667, family holdout 0.054968, combined 0.121542.
- Accepted novelty 0.795501, complexity 0.44, and adaptation efficiency 0.014678.
- Last-epoch runtime 91.2285 ms and distance evaluations 3810.75.
- Validation: surviving False, transplant delta 0.003293, positive scaffolds 1, Pareto runtime inflation 0.197399.
- Validation summary: {"better_gap_same_runtime": false, "code_length_lines": 12, "complexity_score": 0.44, "distance_eval_inflation": 0.154905, "gap_delta": 0.000224, "runtime_inflation": 0.197399, "same_gap_faster": false}

Judge Appendix

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TSP Modular-Operator Discovery Suite - Conservative Interpretation

Summary of Key Metrics to Prioritize

- ****Held-out TSPLIB gap**** and ****family holdout gap**** are main indicators of successful transfer.

- Modular operators must pass ****transplant**** and ****ablation**** checks to be considered reusable.
- No operator here survives scaffold transplant or family holdout validation clearly.
- No operator is marked as a surviving candidate, so ****no algorithm discovery claimed****.

Condition Analysis

1. Baseline (phase7_baseline_heuristic_only)

- Held-out TSPLIB gap: 0.235058
- Family gap: 0.054968
- Transfer gap: 0.121317
- No modular operator.
- Used as baseline reference.

2. Full Solver Evolution (phase7_full_solver_evolution)

- Held-out TSPLIB gap improved to 0.180847 (better than baseline)
- Family gap reduced to 0.0
- Transfer gap: 0.115084 (improvement)
- No modular operator components isolated (operator_type = "n/a").
- No surviving candidate.

3. Modular Operator: det_escape_doublebridge_escalation_perturbation

- Held-out TSPLIB gap: 0.235058 (no improvement over baseline)
- Family gaps: no gains; 0 positive families.
- Ablation: disabling the operator does ****not**** degrade performance (gap delta = 0).
- Transplant test showed no consistent improvement; fails on clustered_local_search.
- Runtime slightly improved or neutral.
- Not a surviving candidate.
- Conclusion: No evidence of reusable operator effect; operator fails transplant and ablation checks.

4. Modular Operator: candidate_prune_transfer_adaptive_local_trap_guard and candidate_prune_transfer_prone_to_clusters_and_grids (similar candidate pruner variants)

- Held-out TSPLIB gap unchanged or slightly worse (0.235667 vs baseline 0.235058).
- Family gap shows no positive families.
- Ablation: disabling operator does ****not**** hurt performance (negative or zero gap delta).
- Transplant: slight positive scaffold count but mean transplant gap delta positive (worse).
- Runtime inflation notable (~0.61 in worst cases).
- Conclusion: No confirmed reuse or positive operator effect; ablation and transplant checks fail.

5. Modular Operator: restart_controller_adaptive_2x_pool and det_restart_controller_two_cluster_bottleneck

- Family holdout gap shows some small improvements on bottleneck/clustered TSP families.
- Ablation: disabling operator ****hurts**** performance by about 0.005gap, indicating a modest operator effect.
- Transplant results: weak positive scaffold counts, but mean transplant gap delta positive (slightly worse).
- Runtime inflation high (~0.62).
- Does not transplant cleanly to some scaffolds.
- No surviving candidate status.
- Conclusion: borderline operator effect but fails robust transplant checks and shows runtime cost; no reliable reusable operator identified.

Pareto Tradeoffs

- Pareto results for all modular operators show:
- No major gap improvements over baseline.
- Some operators slightly faster without gap improvements (e.g., det_escape_doublebridge_escalation_perturbation).
- Others have significant runtime inflation without gap gains.
- No operator dominates on the Pareto front in gap/runtime tradeoff.

Overall Conclusion

- ****No modular operator passes transplant and ablation validation to confirm reusable operator effect.****
- Held-out TSPLIB and family gaps do not improve with modular operators relative to baseline or full solver condition.
- Ablation tests consistently show disabling the operators does not hurt or yields negligible impact.
- Transplant tests fail or show insignificant/negative improvement.
- No surviving candidates identified.
- ****No modular operator discovery or reusable operator structure adoption is supported by the current results.****
- The full solver evolution shows performance gains but lacks modular operator isolation; no modular discovery claim can be made.

Final recommendation:

- Continue refining modular operator designs to produce consistent family holdout gains and robust transplant survival.
- Prioritize operators that demonstrate genuine ablation sensitivity and transfer benefits before claiming reusable operator structures.